

Lists and indexing

Project: Python readiness portfolio

1. Your mission

Your learning target

I can distinguish a list index from the value stored at that position.

I can read and update an item using a valid index.

I can trace a loop through list values and test an empty list.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Distinguish a list index from the value stored at that position

Read and update an item using a valid index

Trace a loop through list values and test an empty list

Retrieve what you already know

What values does `range(3)` supply? How many iterations occur?

My starting explanation

This week's end goal

Trace lists and indexing and explain the changed program before running it.

2. Notice the evidence

A fictional supply shelf has three trays with counts 4, 7 and 4 in that order. Store them as `counts = [4, 7, 4]`. Square brackets enclose the list and commas separate its items. The order matters: the first and third trays contain equal values but occupy different positions.

Python counts positions from zero. `counts[0]` reads the first item, `counts[1]` the second and `counts[2]` the third. `len(counts)` is 3 because there are three items. It is a count, not the last nonnegative index. This lesson uses integer counts and nonnegative indices; Python also supports negative indices, which are outside our current indexing convention.

List the given facts and one fact that is not supplied.

3. See how it works

Positions and stored counts

Tray position · Index · Value

Tray position: First

Index: 0

Value: 4

Tray position: Second

Index: 1

Value: 7

Tray position: Third

Index: 2

Value: 4

Equal stored values do not imply equal positions. There is no item at index 3.

```
counts = [4, 7, 4]
print(len(counts))
print(counts[0])
print(counts[1])
counts[1] = 9
print(counts)
```

The printed lines are 3, 4, 7 and [4, 9, 4]. The assignment `counts[1] = 9` replaces the item at index 1; it does not add a fourth item or change the two other values. The length stays 3. A recorded count can be changed by code without proving that the physical tray really contains that many objects.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Start with `counts = [4, 7, 4]`. Run `counts[0] = 6`. Write the resulting list, its length and the value at index 2. Predict before running.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Replace the second print with `print(counts[3])`.

Prediction: The first print shows 6, then `IndexError` stops the script before the length print.

Reason: The attempted position does not exist.

5. Words that unlock the idea

Word · Meaning

Word: List

Meaning: An ordered collection of items

Word: Index

Meaning: An integer position used to access an item

Word: Length

Meaning: The number of items in a collection

Word: Item

Meaning: One value stored in a collection

Word: `IndexError`

Meaning: An error raised when an index is outside a sequence

Use two terms to explain a step in this week's example.

6. Follow a worked example

```
counts = [4, 7, 4]
```

```
total = 0

for count in counts:
    total = total + count

print(total)
```

Item read as count · total before · total after

Item read as count: 4

total before: 0

total after: 4

Item read as count: 7

total before: 4

total after: 11

Item read as count: 4

total before: 11

total after: 15

This fresh script prints 15. `for count in counts` supplies the stored values, not the indices. Both copies of 4 are processed because both are items. For an empty list written `[]`, the body runs zero times and total stays 0. For `[7]`, it runs once and prints 7.

Change only the list to `[2, 0, 5]`. Trace the three additions and state the final output. Explain why the middle item is processed even though its value is zero.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time.

Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Lists and indexing

Indexes are positions, not values. A list of shelf counts is read by position.

The facts we will use

Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

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Zero-based list positions

	Index 0	Index 1	Index 2
Stored counts	6	1	8

Read a cell using both its row and column labels.

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.



2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.



3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with `print(counts[3])`.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Draw positions

What happens and why: Indexes 0,1,2 contain 6,1,8 respectively.

Step: 2. Read two positions

What happens and why: Index 0 returns 6; index -1 returns the final value 8.

Step: 3. Check length boundary

What happens and why: Length is 3, but index 3 is beyond the final valid positive index 2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

List positions begin at zero. A negative index counts from the end; the list length counts entries and is not itself a valid positive index for the final entry.

What this example does not establish

A valid position does not establish that the stored count is accurate; empty lists also have no final element to read.

Read the labelled picture: Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.

Change one thing: Replace the second print with `print(counts[3])`. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

```
counts = [4, 7, 4]
print(counts[3])
```

This deliberately faulty script raises `IndexError`. The list has three items, but its nonnegative indices are 0, 1 and 2. Replace 3 with 2 to read the third item, which is 4. For any nonempty list, `len(counts) - 1` identifies the last item using a nonnegative index. An empty list has no valid item index.

Avoid confusing values with positions: for `count` in `counts` followed by `counts[count]` would try indices 4, 7 and 4 in this example. If you need both, use `for index in range(len(counts))` and read `counts[index]`. If you only need a total, looping directly through values is simpler.

Locate the earliest unsupported step

A learner loops with `for count in counts` and tries to read `counts[count]` for `counts = [4, 7, 4]`. Identify the first attempted index and error. Explain how to print each value directly.

State one repair and the checks you would rerun.

8. Practise together

For counts = [4, 7, 4], give the length, valid nonnegative indices and value at index 1. Explain how the two items with value 4 differ.

Change only the list to [2, 0, 5]. Trace the three additions and state the final output. Explain why the middle item is processed even though its value is zero.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace an update

Use workbook W2 for the connected example “Lists and indexing”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case.

Each person writes their own explanation.

Check your reasoning

Use the worked case for Lists and indexing. Proposed explanation: “Length is the last nonnegative index”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Label the three index positions. Predict all original outputs, then name the first output and stopping point after using index 3.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Start with `counts = [4, 7, 4]`. Run `counts[0] = 6`. Write the resulting list, its length and the value at index 2. Predict before running.

```
counts = [4, 7, 4]
print(len(counts))
print(counts[0])
print(counts[1])
counts[1] = 9
print(counts)
```

```
counts = [4, 7, 4]
total = 0
for count in counts:
    total = total + count
```

```
print (total)
```

```
counts = [4, 7, 4]  
print (counts[3])
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For `counts = [4, 7, 4]`, give the length, valid nonnegative indices and value at index 1. Explain how the two items with value 4 differ.

A learner loops with `for count in counts` and tries to read `counts[count]` for `counts = [4, 7, 4]`. Identify the first attempted index and error. Explain how to print each value directly.

Compare `[1, 5, 2]` and `[5, 1, 2]`. State their totals and first items. Would storing only the total answer which first item was larger? Explain.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled lists and indexing evidence sheet to Python readiness portfolio. Show how you can distinguish a list index from the value stored at that position. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

12. Show what you know

Before running a script, sketch positions above the stored values. Test an empty list, a one-item list and a list with repeated values. These cases reveal different mistakes. A total alone loses order: [4, 7, 4] and [7, 4, 4] both total 15 even though their first items differ. Data checks should retain the detail needed for the question.

A learner uses `counts[len(counts)]` to read the final item. Explain the error for [8, 3] and give a valid nonnegative index. Explain why your repair still needs an empty-list check.

Compare [1, 5, 2] and [5, 1, 2]. State their totals and first items. Would storing only the total answer which first item was larger? Explain.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Positions and stored counts

Tray position · Index · Value

Tray position: First

Index: 0

Value: 4

Tray position: Second

Index: 1

Value: 7

Tray position: Third

Index: 2

Value: 4

Equal stored values do not imply equal positions. There is no item at index 3.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Distinguish a value from an index

Predict the three outputs. If the second print uses values[3] instead, where does execution stop and which later output is not produced?

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Start with readings = [3, 0, 3, 2]. Replace the second item with 5 using its index. Write a loop that adds all updated readings and prints one final total. Trace every update of the total. Then test the summing part alone on [] and [6].

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Indexes are positions, not values. A list of shelf counts is read by position.

By the end, I can · How I will show it

By the end, I can: Distinguish a list index from the value stored at that position

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Read and update an item using a valid index

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Trace a loop through list values and test an empty list

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Distinguish a list index from the value stored at that position

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.